EXPNO: 07
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Sliding Window Protocol

Aim:

To implement flow control at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow frames from one node to another.

Sender Program:

1. Input window size from the user
2. Input a test message from the user
3. Consider 1 character per frame.
4. Create a frame with following fields
   [Frame no., DATA]
5. Send the frames.
6. Wait for the acknowledgment from the Receiver.
7. Reader a file called Receiver_Buffer.
8. Check ACK field for the Acknowledgement no.
9. If the Acknowledgement no is as expected, send new set of frames accordingly, Else if NACK is received, resend the frames accordingly.

Create a receiver file with following:

1. Reads a file called Sender_Buffer
2. Check the Frame no
3. If the Frame no are as expected, write the appropriate ACK in the Receiver_Buffer file. Else write NACK no. in the Receiver_Buffer file.

Code:

```
import time
import os
sender_file = "Sender_Buffer.txt"
receiver_file = "Receiver_Buffer.txt"
message = input ("Enter text message: ")
window_size = int (input ("Enter window
                          size :"))
frames = [ ]
for i, ch in enumerate (message):
    frames.append ((i+1, ch))
start = 0
while start < len (frames):
    end = min (start + window_size,
               len (frames))
    with open (sender_file, "w") as f:
        for i in range (start, end):
            print (f "FrameNo: {frames[i][0]}
                   DATA: {frames[i][1]}")
```

```
print ("\n Waiting for acknowledgement...")
time.sleep(2)

if not os.path.exists (receiver_file):
    print ("Receiver buffer not found.")
    break
with open (receiver_file, "r") as f:
    ack = f.read().strip()
print ("Receiver Response:", ack)
if ack.startswith ("ACK"):
    ack_no = int (ack.split (":") [1])
    if ack_no == end:
        print ("Acknowledgement recieved.")
        start = end
    else:
        print ("Unexpected Ack")
elif ack.startswith ("NACK"):
    nack_no = int (ack.split (":") [1])
    print (f"NACK recieved for Frame")
    print ("Resending frames...")
    start = nack_no - 1
print ("\n All frames transmitted")
```

Receiver program:

```
with open ("sender_Buffer.txt") as f:
    frames = f.readlines()
expected = 1
```

```
for frame in frames:
    no, data = frame.strip().split(",")
    if int(no) == expected:
        print("Frame", no, "accepted")
        expected += 1
    else:
        print("NACK:", expected)
        break
else:
    with open("Receiver Buffer.txt", "w") as f:
        f.write("ACK:" + str(expected-1))
    print("ACK", expected-1)
```

Input :-

Enter the text message : HELLO

Enter Window size : 3

output (Sender):-

Frames sent:

Frame No: 1 DATA:H

Frame No: 2 DATA:E

Frame No: 3 DATA:L

Waiting for Acknowledgement...

Receiver Response: ACK : 3

Acknowledgement recieved correctly

Output - Reciever

Recieved Frame No : 1 DATA : H

Frame accepted

Recieved Frame No: 2 DATA : E

Frame accepted

Recieved Frame No: 3 DATA : L

Frame accepted.

Result :

Thus, Sliding Window is implemented successfully.